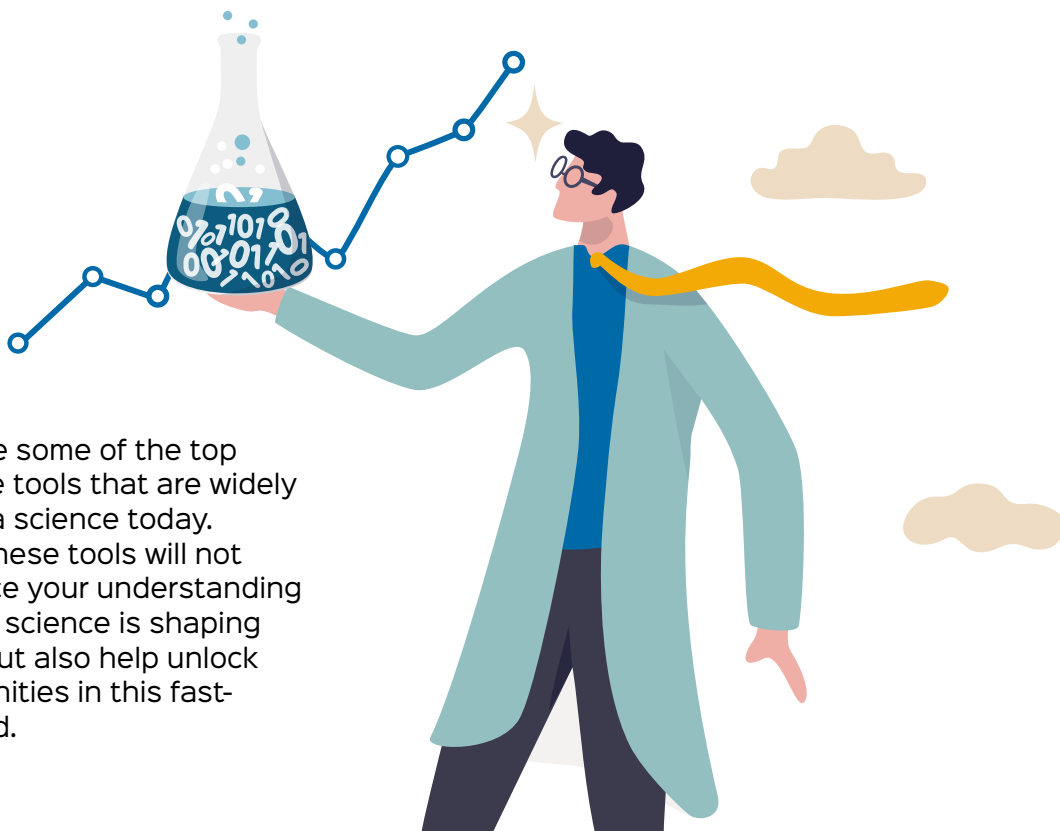


# Essential Open Source Tools for Budding Data Scientists



Let's explore some of the top open source tools that are widely used in data science today. Mastering these tools will not only enhance your understanding of how data science is shaping industries but also help unlock job opportunities in this fast-growing field.

**D**ata science has today emerged as a very important field in the technology industry and is helping companies use data for better decisions and profits. According to *Fortune Business Insights*, the global data science platform market was valued at around US\$ 104 billion in 2023 and is expected to grow to nearly US\$ 777 billion by 2032, with a compound annual growth rate of nearly 25%. This rapid growth of data science has happened because of the increasing availability of data, improvements in computational power and the development of sophisticated machine learning algorithms.

While most organisations use proprietary tools for their large scale data science operations, open source platforms like Python, R, and Apache Spark are popular not just in academia but across various industries. These tools are free and anyone can use, modify and contribute to their development. As businesses increasingly seek professionals who can navigate data analysis, machine learning, and business intelligence, knowledge of open source platforms like Python and R makes a candidate more attractive to employers.

Many leading companies today actively contribute to the open source

ecosystem even while primarily depending on proprietary software. For example, Google has released TensorFlow, a widely used open source machine learning library that allows anyone to create and deploy AI models. Facebook has developed PyTorch, another popular open source tool for machine learning and deep learning. Microsoft has also contributed significantly by releasing open source tools like Visual Studio Code, a coding platform loved by developers worldwide. Apache Spark, developed at UC Berkeley, is another powerful tool used to process large amounts of data quickly and efficiently. Big companies